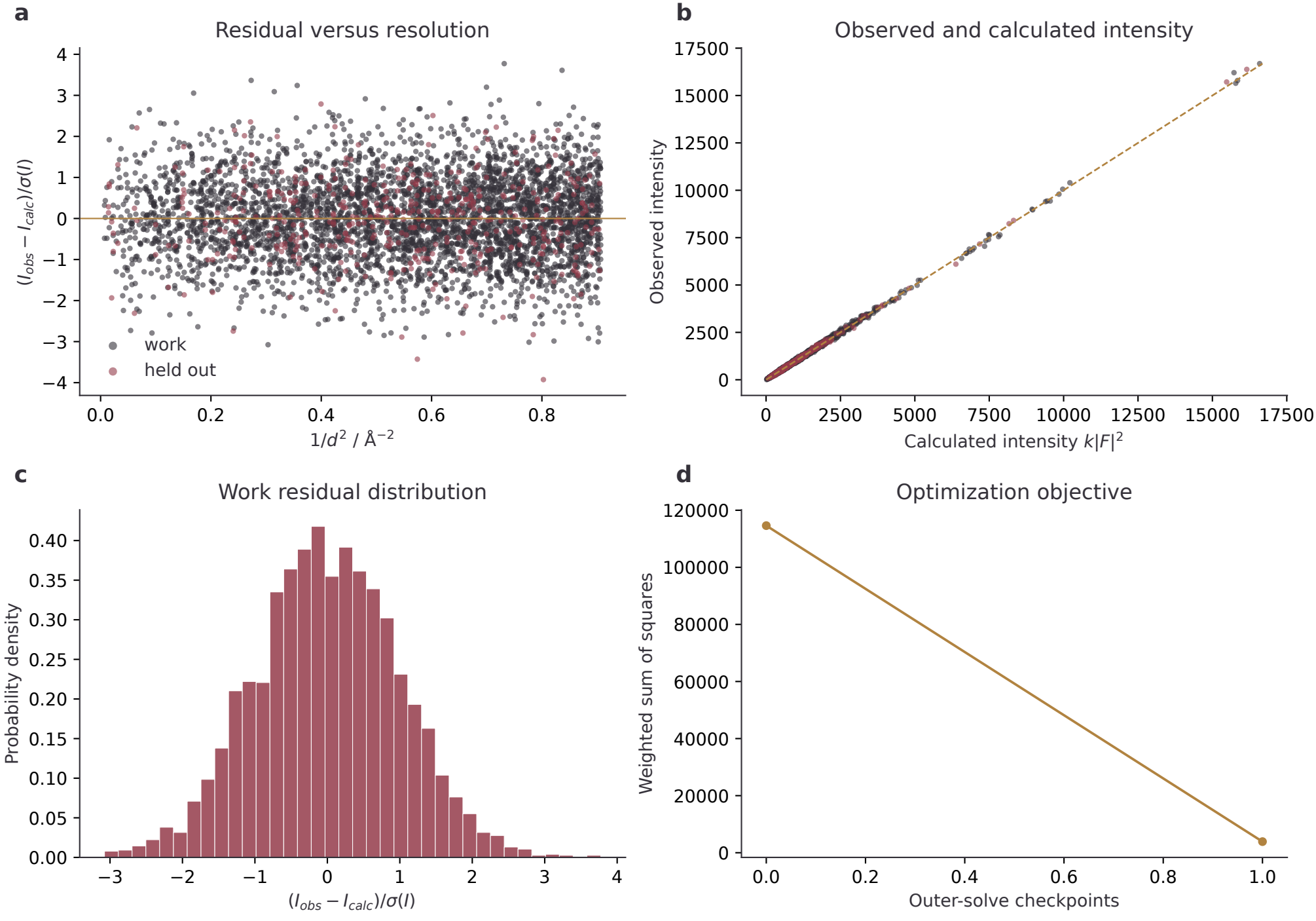

a
Residual versus resolution
$(I_{obs} - I_{calc})/\sigma(I)$
$1/d^2$ / Å^{-2}
work
held out
b
Observed and calculated intensity
Observed intensity
Calculated intensity $k|F|^2$
c
Work residual distribution
Probability density
$(I_{obs} - I_{calc})/\sigma(I)$
d
Optimization objective
Weighted sum of squares
Outer-solve checkpoints